

1. Business Understanding (Domain Exploration)

Core Business Processes

The beauty salon operates by providing personal care services to customers through scheduled appointments. The main process begins when a customer books an appointment for a specific service and employee. The employee performs the service at the scheduled time, after which the customer makes a payment.

The system must support the full lifecycle of salon operations: appointment booking, service delivery, payment processing, and record keeping.

Main services offered:

- Haircuts
- Hair coloring
- Manicures and pedicures
- Facial treatments
- Hair styling

Pain Points Without Software:

- Manual appointment scheduling may lead to overlapping bookings.
- Customer information is difficult to track and update.
- Service history for customers is not easily accessible.
- Payments and revenue are recorded manually, increasing the risk of errors.
- Customers may forget appointments because reminders are not sent automatically.
- It is difficult for the owner to analyze business performance.

Goals of the System:

- Digitize appointment scheduling
- Store and manage customer information
- Track services and employee schedules
- Record payments and billing information
- Send appointment reminders to customers
- Provide business reports and analytics

- Improve operational efficiency and customer satisfaction
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2. Functional Features

Customer Management

The system will allow staff to create and manage customer profiles.

Features:

- Registering new customers
- Editing customer information
- Viewing service history
- Searching for customers

Appointment Scheduling

The system will allow staff to create and manage appointments.

Features:

- Booking appointments for customers
- Assigning services and employees
- Viewing daily and weekly schedules
- Rescheduling or canceling appointments
- Preventing scheduling conflicts

Staff Management

The system will store employee information and manage their work schedules.

Features:

- Adding and editing employees
- Assigning services employees can perform
- Managing employee schedules
- Tracking employee performance

Payments & Billing

The system will record payments after services are completed.

Features:

- Recording payment for completed services
- Tracking revenue
- Supporting different payment methods (cash, card, digital payment)
- Generating receipts

Notifications

To reduce missed appointments, the system will support notifications.

Examples:

- Appointment reminders sent to customers
- Appointment confirmation messages
- Notifications for schedule changes

Reports

The system will generate useful business reports for the owner.

Examples:

- Daily revenue report
- Monthly revenue summary
- Most popular services
- Employee performance statistics
- Number of appointments per period

3. Domain Model (Conceptual)

Main Entities / Aggregates

Customer

Represents a salon client who books services.

Employee

Represents a staff member who performs salon services.

Service

Represents a treatment offered by the salon.

Appointment

Represents a scheduled meeting between a customer and employee for a specific service.

Payment

Represents a completed financial transaction for an appointment.

Relationships

Customer → Appointment

A customer can have multiple appointments.

Employee → Appointment

An employee performs multiple appointments.

Service → Appointment

Each appointment includes one or more services.

Appointment → Payment

Each completed appointment generates a payment record.

Important Business Rules:

- An employee cannot have two appointments at the same time.
- Each appointment must be linked to a customer and at least one service.
- Payment is recorded only after the service is completed.
- Only employees qualified for a service can perform it.
- Appointment duration is determined by the selected service.

Value Objects:

ServicePrice

Represents the price of a service.

AppointmentTime

Represents the start time and duration of an appointment.

ContactInformation

Represents phone number and email information for customers.

4. Database Structure (High-Level)

Customers

Fields:

- customer_id (PK)
- name
- phone
- email
- created_at

Employees

Fields:

- employee_id (PK)
- name
- role
- phone
- email

Services

Fields:

- service_id (PK)
- name
- description
- price
- duration

Appointments

Fields:

- appointment_id (PK)
- customer_id (FK)
- employee_id (FK)

- appointment_date
- start_time
- status

Payments

Fields:

- payment_id (PK)
- appointment_id (FK)
- amount
- payment_method
- payment_date

5. User Roles & Permissions

Owner / Administrator

Permissions:

- View all data
- Create, edit, and delete services
- Manage employees
- View financial reports
- Approve or cancel appointments

Staff Member

Permissions:

- View their own appointments
- Update appointment status
- View customer information
- Record completed services

Receptionist

Permissions:

- Create appointments
- Edit or cancel appointments
- Register new customers
- View schedules

Customer

Permissions:

- Book appointments online
- View their appointments
- Cancel or reschedule bookings
- Receive notifications